

Geometric Interpretation of Differentiation

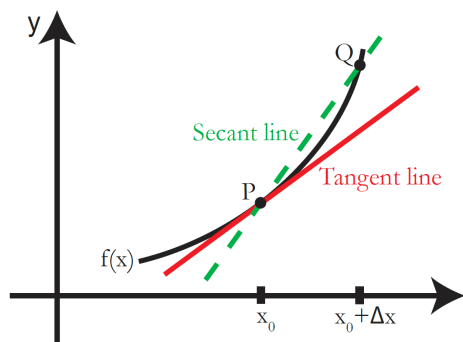


Figure 1: A graph with secant and tangent lines.

The derivative of $f(x)$ at $x = x_0$ is the slope of the tangent line to the graph of $f(x)$ at the point $(x_0, f(x_0))$. But what is a tangent line?

- It is NOT just a line that meets the graph at one point.
- It is the *limit* of the secant lines joining points $P = (x_0, f(x_0))$ and Q on the graph of $f(x)$ as Q approaches P .

The tangent line touches the graph at $(x_0, f(x_0))$; the slope of the tangent line matches the direction of the graph at that point. The tangent line is the straight line that best approximates the graph at that point.

Given a graph of our function, it's not hard for us to draw the tangent line to the graph. However, we'll want to do computations involving the tangent line and so will need a computational method of finding the tangent line.

How do we compute the equation of the line tangent to the graph of the function $f(x)$ at a point $P = (x_0, y_0)$? We know that the equation of the straight line with slope m through the point (x_0, y_0) is $y - y_0 = m(x - x_0)$, so in the abstract we know the equation of the tangent line.

To get a specific equation for the line, we'll need to know the coordinates x_0 and y_0 of the point P . If we know x_0 we can find $y_0 = f(x_0)$ by substituting the value x_0 into the expression for $f(x)$. The second thing we need to know is the slope, $m = f'(x_0)$, which we call *the derivative of f* .

Definition: The derivative $f'(x_0)$ of f at x_0 is the slope of the tangent line to $y = f(x)$ at the point $P = (x_0, f(x_0))$.

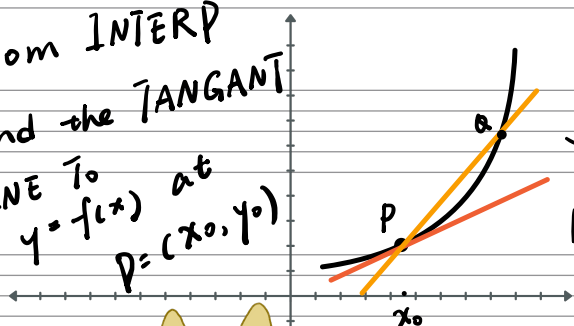
A *secant* line is a line that joins two points on a curve. If the two points are close enough together, the slope of the secant line is close to the slope of the curve. We want to find the slope of the tangent line m — which equals the slope of the curve — and we use the slopes of secant lines to do this.

Suppose PQ is a secant line of the graph of $f(x)$. We can find the slope of the graph at P by calculating the slope of PQ as Q moves closer and closer to P (and the slope of PQ gets closer and closer to m).

The tangent line equals the limit of secant lines PQ as $Q \rightarrow P$; here P is fixed and Q varies.

Geom INTERP

Find the TANGENT LINE to $y = f(x)$ at $P = (x_0, y_0)$



$$y - y_0 = m(x - x_0) \quad \text{slope } m = f'(x)$$

$$\text{point } y_0 = f(x_0)$$

DEF'N $f'(x)$, the derivative of f at x_0 ,

is the slope of the tangent line to $y = f(x)$ at P .

TANGENT LINE = LIMIT of Secant LINES PQ as $Q \rightarrow P$

Slope as Ratio

While we're still thinking geometrically, we can now use symbols and formulas in our computation.

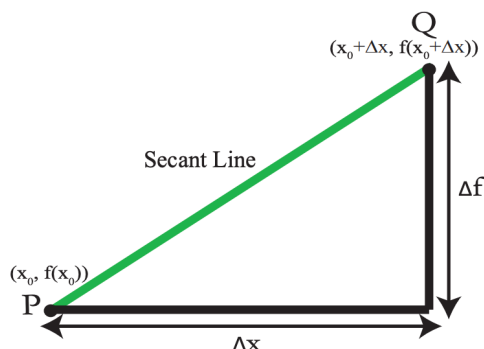


Figure 1: Geometric definition of the derivative

We start with a point $P = (x_0, f(x_0))$. We move over a tiny horizontal distance Δx (pronounced "delta x " and also called "the change in x ") and find point $Q = (x_0 + \Delta x, f(x_0 + \Delta x))$. These two points lie on a secant line of the graph of $f(x)$; we will compute the slope of this line. The vertical difference between P and Q is $\Delta f = f(x_0 + \Delta x) - f(x_0)$.

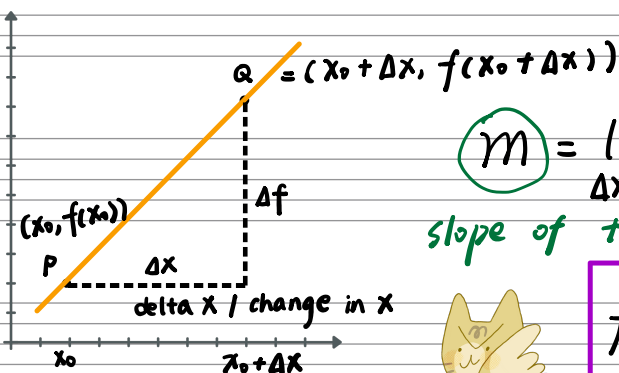
The slope of the secant PQ is rise divided by run, or the ratio $\frac{\Delta f}{\Delta x}$. We've said that the tangent line is the limit of the secant lines. It is also true that the slope of the tangent line is the limit of the slopes of the secant lines. In other words,

$$m = \lim_{Q \rightarrow P} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

We started with a point P on the graph of $y = f(x)$ which had coordinates $(x_0, f(x_0))$. We then found a point Q on the graph which was Δx units to the right of P . The coordinates of Q must be $(x_0 + \Delta x, f(x_0 + \Delta x))$. We can now write the following formula for the derivative:

$$m = \underbrace{f'(x_0)}_{\text{derivative of } f \text{ at } x_0} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \underbrace{\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}}_{\text{difference quotient}}$$

This is by far the most important formula in Lecture 1; it is the formula that we use to compute the derivative $f'(x_0)$, which equals the slope of the tangent line to the graph at P . A machine could use this formula together with the coordinates $(x_0, f(x_0))$ of the point P to draw the tangent line to the graph of $y = f(x)$ at the point P .



$$m = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} \quad \text{slope of secant } PQ$$

slope of tangent

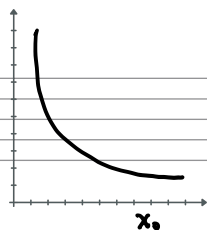


$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

(= m)

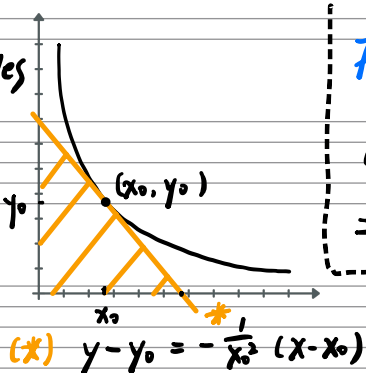
difference quotient

Example 1 $f(x) = \frac{1}{x}$ $f'(x_0) = -\frac{1}{x_0^2}$ < 0 negative slope, $x_0 \rightarrow \infty$, less & less steep.



$$\begin{aligned} \frac{\Delta f}{\Delta x} &= \frac{\frac{1}{x_0 + \Delta x} - \frac{1}{x_0}}{\Delta x} = \frac{1}{\Delta x} \left(\frac{x_0 - (x_0 + \Delta x)}{(x_0 + \Delta x)x_0} \right) = \frac{1}{\Delta x} \left(\frac{-\Delta x}{(x_0 + \Delta x)x_0} \right) \\ &= \frac{-1}{(x_0 + \Delta x)x_0} \xrightarrow{\Delta x \rightarrow 0} \boxed{-\frac{1}{x_0^2}} \end{aligned}$$

Find Areas of Triangles enclosed by AXES and TANGENT to $y = \frac{1}{x}$



Area of Δ :

$$\frac{1}{2} (2x_0) (2y_0) = 2x_0 y_0 = 2$$

Find x-intercept ($y=0$)

$$0 - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0) = -\frac{x}{x_0^2} + \frac{1}{x_0}$$

$$\Rightarrow \frac{x}{x_0^2} = \frac{2}{x_0} \Rightarrow \boxed{x = 2x_0}$$

Shortcut to y-Intercept use symmetry $y = 2y_0$

$(x, y) \rightarrow (y, x)$

Symmetry explanation

$$y = \frac{1}{x} \Leftrightarrow xy = 1 \Leftrightarrow x = \frac{1}{y}$$

Can also get y-intercept by plugging $x=0$ into (*)

More Notations $f' = \frac{df}{dx} = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}y$

$y=f(x)$, $\Delta y = \Delta f$ Newton's Leibniz

(omits x_0)

Example 2. $f(x) = x^n$, $n = 1, 2, 3, \dots$ (x is fixed
 Δx moves)

d $\frac{d}{dx} x^n = ?$ $\frac{\Delta f}{\Delta x} = \frac{(x+\Delta x)^n - x^n}{\Delta x}$

$\frac{d}{dx} x^n = nx^{n-1}$ Extends to polynomials

$\frac{d}{dx} (x^3 + 5x^{10}) = 3x^2 + 50x^9$



(binomial theorem)

$(x + \Delta x)^n = (x + \Delta x) \cdots (x + \Delta x)$

$= x^n + nx^{n-1}\Delta x + \text{junk } \underline{O((\Delta x)^2)}$ terms $\psi / (\Delta x)^2, (\Delta x)^3 \dots$
big O of delta x squared

$\frac{\Delta f}{\Delta x} = \frac{1}{\Delta x} ((x + \Delta x)^n - x^n)$

$= \frac{1}{\Delta x} (x^n + nx^{n-1}\Delta x + O((\Delta x)^2) - x^n)$

$= \frac{1}{\Delta x} (nx^{n-1}\Delta x + O((\Delta x)^2))$

$= nx^{n-1} + O(\Delta x)$

$\xrightarrow{\Delta x \rightarrow 0} nx^{n-1}$